SPHEREx Discovery Engine: System Architecture & Scientific Guide

NASA Space Apps Challenge 2026 | Time-Domain All-Sky Survey, 102 Spectral Channels & Citizen Science Hunter

1. Executive Overview & NASA Mission Context

NASA's Spectro-Photometer for the History of the Universe, Epoch of Reionization, and Ices Explorer (**SPHEREx**), launched into a Sun-synchronous low Earth orbit in early 2025, surveys the entire sky once every 6 months across **102 contiguous near-infrared spectral bands (0.75 μm to 5.00 μm)**. Because SPHEREx repeats all-sky coverage every 6 months, objects within our solar system and solar neighborhood exhibit prominent apparent motion caused by a combination of intrinsic Keplerian orbital motion, stellar proper motion, and annual Earth parallax.

The **SPHEREx Discovery Engine** is a zero-dependency, high-performance web application engineered specifically to solve this data avalanche. It empowers both professional astrophysicists and citizen scientists to rapidly identify moving objects (comets, asteroids, Kuiper Belt objects, cold brown dwarfs) and variable stars (T Tauri and Mira pulsators) using automated blinking, pixel-level flux subtraction, 6-detector coaddition, and on-the-fly 102-band spectral energy distribution (SED) extraction.

2. NASA Challenge Benchmark Targets & Physical Characteristics

Target Name	Equatorial Coordinates	Astrophysical Class	Motion / Variability Diagnostics in Tool
Barnard's Star (GJ 699)	RA: 17h 57m 48.5s Dec: +04° 41' 36"	High Proper-Motion M4V Red Dwarf	Record proper motion of **10.37 "/yr**. In the 6-month baseline, shifts 5.18" (10.4 px), creating a distinct spatial dipole in Subtract mode.
Epsilon Indi B (Ba / Bb)	RA: 22h 04m 10.5s Dec: -56° 46' 58"	Binary Brown Dwarfs (T1 + T6)	Fast proper motion of **4.70 "/yr**. The 102-band SED reveals deep 1.6 μm and 3.3 μm methane (CH4) absorption bands.
WISE 0855-0714 (Sub-stellar Cold)	RA: 08h 55m 10.83s Dec: -07° 14' 42.5"	Coldest Known Y-Dwarf (250 K)	Proper motion of **8.15 "/yr**. Near-zero flux in SWIR (0.75-2.0 μm); emits almost entirely in thermal MWIR (4.0-5.0 μm) water-ice bands.
RW Aurigae (RW Aur A/B)	RA: 05h 07m 49.5s Dec: +30° 24' 05"	T Tauri Young Variable Star	**Stationary position (0 px shift)**, but flux fluctuates by 3-4 magnitudes due to circumstellar disk dust occultation; forms an amplitude residual peak.
R Draconis (Mira Variable)	RA: 16h 32m 40.2s Dec: +66° 45' 18"	Pulsating AGB Red Giant	**Stationary position**, but exhibits periodic 245-day thermal pulsation and carbon/water envelope variations across 2-5 μm.
Real SPHEREx QR2 (2026W08_ZA_0157)	RA: 04h 12m 30.0s Dec: +18° 35' 00"	Calibrated NASA Level-2 FITS	Direct calibrated survey field with 0.75-5.00 μm calibrated pixel arrays in MJy/sr with zero cosmic-ray/lunar contamination.

3. Time-Domain Comparison & Signal Processing Pipeline

Mode	Mathematical Operation	Astrophysical Purpose
Blink Mode	$I_{disp}(x,y,t) = I_{E1}$ if $(t \bmod 2T < T)$ else I_{E2}	Alternates between epochs every 0.15s to 1.50s. Leverages human visual persistence to immediately catch moving sources.
Subtract (ΔI)	$\Delta I(x,y) = I_{E2}(x,y) - I_{E1}(x,y) + \text{Offset}$	Cancels all stationary background stars and galaxies to zero. Moving sources appear as a **negative pit (Epoch 1) and positive peak (Epoch 2) dipole**.
Add / Coadd	$I_{coadd}(x,y) = \Sigma I_k(x,y) / N$	Stacks multi-epoch and multi-detector frames. Uncorrelated background noise decreases as $1/\sqrt{N}$, boosting detection significance.
6-Detector Array Coadd	$I_{tot}(x,y) = \Sigma_{d=1..6} I_d(x,y) / 6$	Coadds all 6 SPHEREx detector focal planes (Det 1-3 SWIR, Det 4-6 MWIR), boosting point-source S/N by up to $\sqrt{6} \approx$ **2.45×**.

4. Advanced NASA Requirements Implemented

• **URL Query String Deep Linking:** The tool can be directly embedded or shared via URL parameters: `index.html?target=wise0855&ra=08h55m10.83s&dec=-07d14m42.5s&fov=30`. Clicking 'Share URL' automatically copies the exact view link.
• **Multi-Mission Overlays:** Users can toggle **Gaia DR3** proper motion velocity vectors and **NASA allWISE** infrared contour references to cross-check cataloged stars.
• **Dynamic Field of View (FOV):** Toggle between **5' (Fine)** for brown dwarf separation, **15' (Standard)**, **30' (Wide)**, and **60' (Survey)** for high-speed trans-Neptunian comets.
• **Quality & Contamination Flags:** Reports NASA pipeline quality flags (e.g. `0x00: QUAL_OK`, Zodiacal light contamination, Moon glare, or saturated pixels).
• **On-the-Fly 102-Band SED Extraction:** Powered by Chart.js, users can scrub across all 102 channels (0.75 to 5.00 μm) and visualize diagnostic molecular absorption lines (CH4 at 1.6 & 3.3 μm, H2O at 1.9 & 2.7 μm, CO at 4.6 μm).
• **Export Certified JSON Dossier:** One-click export of structured astrometric reports containing target coordinates, displacement vector, S/N significance, and full 102-channel flux arrays.

5. Integrated Real NASA Data Sources

1. **NASA/IPAC Infrared Science Archive (IRSA):** Calibrated Level-2 SPHEREx Spectral Images in FITS format via the SIA2 API (`https://irsa.ipac.caltech.edu/SIA`) and AWS S3 open bucket.
2. **NASA WISE / NEOWISE & 2MASS:** All-sky infrared survey baseline layers streaming via Aladin Lite CDS endpoints.
3. **Gaia DR3 Catalog (ESA):** Proper motion astrometry vectors (pmRA, pmDec) for celestial kinematic cross-matching.
4. **CDS SIMBAD & Sesame Name Resolver:** Real-time celestial coordinate verification for deep-space targets.

6. Quick Start & Judge Verification

1. Open in any modern web browser: `file:///D:/nasa/index.html`
2. Test each benchmark target from the dropdown: **Barnard's Star, Epsilon Indi B, WISE 0855, RW Aurigae, R Draconis**.
3. Or enter custom coordinates into the **RA and Dec input boxes** and click **Locate Sky Patch**.
4. Toggle through **Blink, Subtract, Add (S/N), and Split** to inspect time-domain dynamics.
5. Scrub the **102-Band Spectral Slider** at the bottom to watch the infrared spectral changes.
6. Click **Export Report** to download the certified JSON discovery dossier.